Chapter 9, Problem 30P

Problem

A voltage $v(t) = 100\cos(60t + 20°)$ V is applied to a parallel combination of a 40-kΩ resistor and a 50-μF capacitor. Find the steady-state currents through the resistor and the capacitor.

Step-by-step solution

Step 1 of 4

Express the applied voltage $v(t)$ in phasor domain.

$V = 100\angle 20°$ V

Calculate the impedance offered by the capacitor.

$$\begin{aligned}Z_C &= \frac{1}{j\omega C}\\ &= \frac{1}{j(60)(50\times10^{-6})}\\ &= \frac{1}{j(3\times10^{-3})}\\ &= -j\frac{1000}{3}\ \Omega\end{aligned}$$

Draw the equivalent circuit as shown in Figure 1.

Step 2 of 4

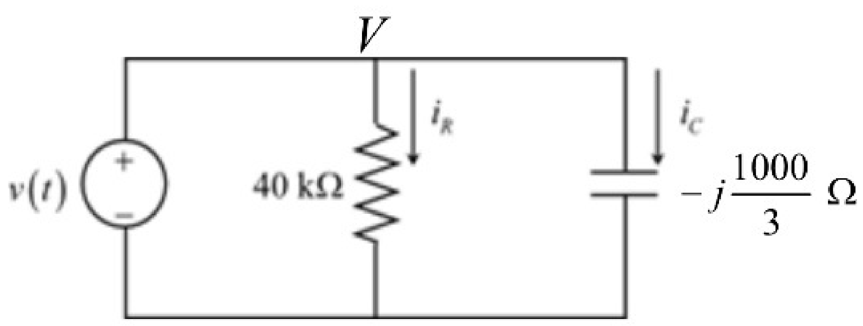

Figure: 1

Step 3 of 4

Determine the current passing through the resistor.

$$\begin{aligned} I_R &= \frac{V}{R} \\ &= \frac{100\angle 20^\circ}{40\times 10^3} \\ &= \left(2.5\times 10^3\right)\angle 20^\circ \\ &= 2.5\angle 20^\circ \text{ mA} \end{aligned}$$

Express the current passing through the resistor in time domain.

$$i_C(t) = 2.5\cos(60t + 20^\circ) \text{ mA}$$

Thus, the steady state current passing through the resistor is $\boxed{2.5\cos(60t + 20^\circ) \text{ mA}}$.

Step 4 of 4

Determine the current passing through the capacitor.

$$\begin{aligned} I_C &= \frac{V}{Z_C} \\ &= \frac{100\angle 20^\circ}{-j\dfrac{1000}{3}} \\ &= \frac{100\angle 20^\circ}{\dfrac{1000}{3}\angle -90^\circ} \\ &= 0.3\angle(90^\circ + 20^\circ) \text{ A} \end{aligned}$$

Express the current passing through the capacitor in time domain.

$$\begin{aligned} i_C(t) &= 0.3\cos(60t + 90^\circ + 20^\circ) \\ &= 0.3\cos(90^\circ + (60t + 20^\circ)) \\ &= -0.3\sin(60t + 20^\circ) \\ &= -300\sin(60t + 20^\circ) \text{ mA} \end{aligned}$$

Thus, the steady state current passing through the capacitor is $\boxed{-300\sin(60t + 20^\circ) \text{ mA}}$.